Name: Vansh Sharma **Roll No:** 2401109 **Subject:** ASDD

Expt No. 6: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

Project Name: Jira-Exp

Project Key: SCRUM-8

Collaborator Email: vanshsharma020406@gmail.com, vinulucifer964@gmail.com

GitHub Repository: https://github.com/HammadShaikh-051/Jira-Experiment

STEP-BY-STEP PROCEDURE

Step 1: Scrum Project Creation s Team Member Invitations

1. Navigate to Atlassian Jira Cloud and log in to the workspace.
2. From the global navigation bar, select **Projects** (or **Spaces**) → **Create project**.
3. Select the **Scrum** methodology template under the *Software Development* category and set the project type to **Team-managed**.
4. Configure project properties: ○ **Project Name: Jira Experiment**
    ○ **Project Key:** SCRUM
5. Click **Create project**.
6. Navigate to the top navigation bar, click the **Add people / Team** icon, enter collaborator email vanshsharma020406@gmail.com, vinulucifer964@gmail.com assign project access permissions as **Member**, and dispatch the invitation.

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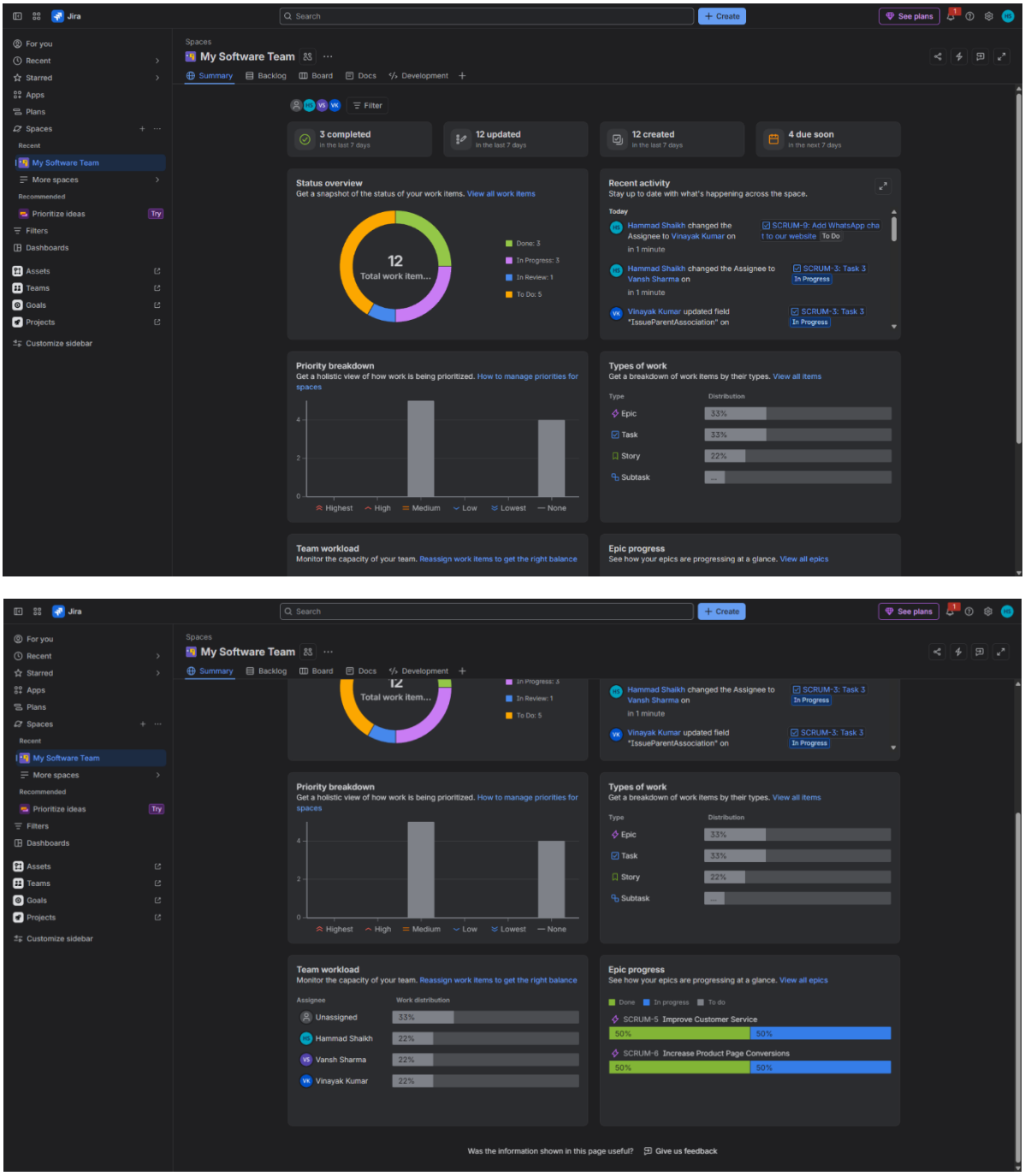


Figure 1: Project Creation view showing project name, key, and team collaborator invitation pane.

Step 2: Epic Definition s Product Backlog Population

1. Open the left sidebar and select the **Backlog** view.
2. Access the **Epics panel** and create the primary delivery epics:

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- **Epic 1:** Improve Customer Service (Colour: Purple) ○ **Epic 2:** Increase Product Page Conversions (Colour: Green)

3. Populate the Product Backlog by clicking **+ Create issue** and creating the core stories and tasks:
   - Scrum-1: Provide live chat as a feature (Story) ○ Scrum-02: Add WhatsApp chat to our website (Task) ○ SCRUM-6: Add Facebook messenger chat to our website (Task) ○ SCRUM-7: Create a video-based knowledge base (Story)
   - SCRUM-8: Create a review-based system for collecting reviews (Story)

4. Open each issue detail view and link them to their respective parent Epics via the **Parent** field.
5. Establish work item dependencies:
   - Open story SCRUM-4 (Provide live chat as a feature). ○ Click **Link issue** (chain icon).
   - Set dependency type to **is blocked by** and select child tasks SCRUM-5 and SCRUM-6.

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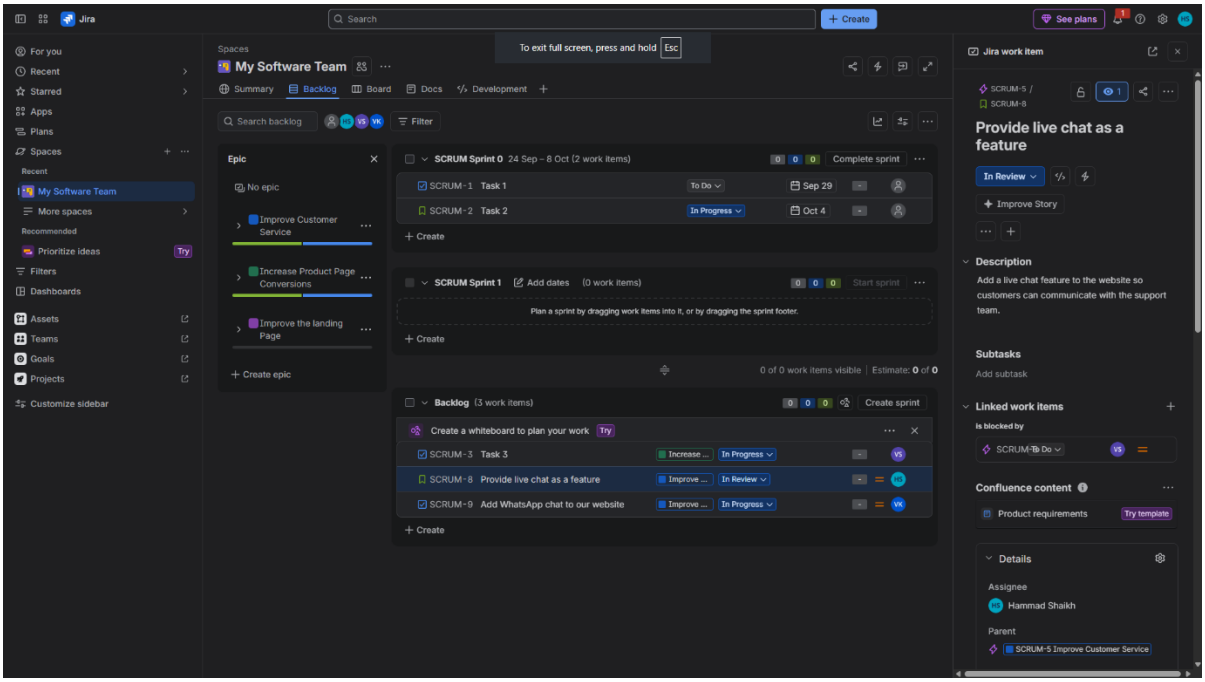


Figure 2: Product Backlog view displaying all items assigned to Epics, alongside the issue detail pane showing the "is blocked by" dependency link.

Step 3: Sprint Planning, Sprint Backlog s Board Customization

1. In the Backlog view, select the active sprint container (SCRUM Sprint 1 / Scrum Sprint 1).
2. Click the three dots (···) on the container header → **Edit sprint**, and define the sprint scope:
    - **Sprint Name:** Scrum Sprint 1
    - **Duration:** Custom
    - **Sprint Goal:** Improving our customers experience on our website
3. Drag SCRUM-4, SCRUM-5, and SCRUM-6 into the sprint container (forming the **Sprint Backlog**), leaving SCRUM-7 and SCRUM-8 in the lower container (remaining **Product Backlog**).
4. Click **Start sprint** to initialize the live Scrum board.
5. In the **Board** view, click the + icon on the far right column header.
6. Create a custom status column named **Review** and position it between **In Progress** and **Done**.

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7. Transition task SCRUM-5 to **In Progress** and SCRUM-6 to **Review** to model workflow progression.

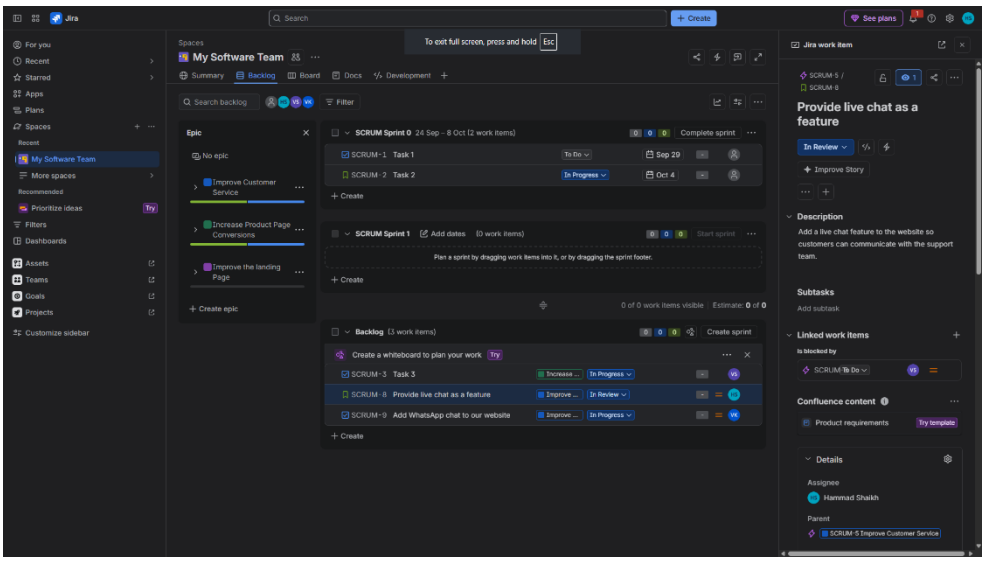


Figure 3A: Jira Backlog screen illustrating the populated Sprint Backlog container separated from the Product Backlog below.

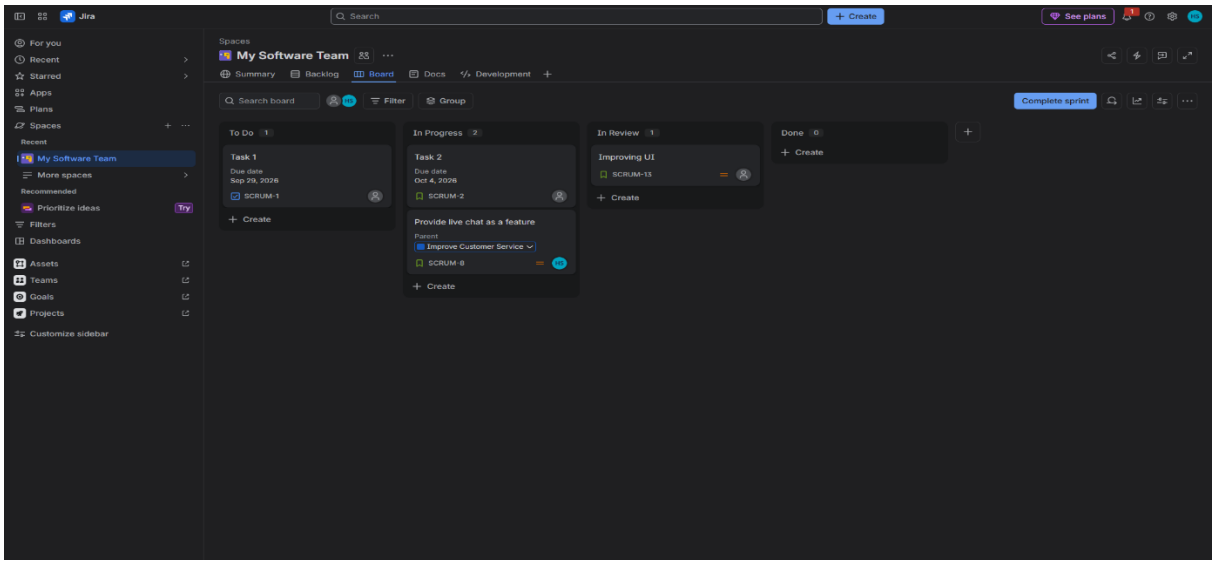


Figure 3B: Active Scrum Board showing the 4-stage workflow pipeline: To Do | In Progress | Review | Done with active issue cards.

Step 4: GitHub Integration with Jira Cloud

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1. In Jira, navigate to **Apps** → **Explore more apps** (or open the </> Development / </> Code tab in the project header).
2. Install and SCRUMch the official **GitHub for Jira** application.
3. Select **GitHub Cloud**, complete OAuth authorization via Atlassian, and connect the profile HammadShaikh-51
4. Under repository selection, grant access permissions specifically to the repository: HammadShaikh-51/Jira-Experiment
5. Confirm that the repository displays an active green **Connected** status inside Jira.

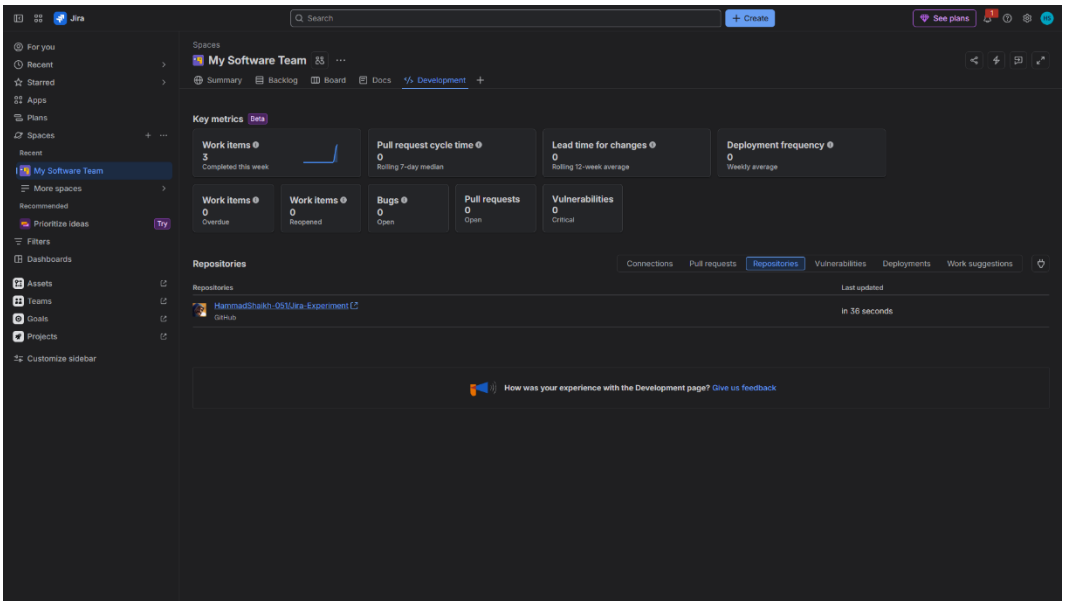


Figure 4: GitHub for Jira connection management page confirming successful linkage of repository HammadShaikh-51/Jira-Experiment

Step 5: Code Commit s Automated Smart Commit Synchronization

1. Open the linked GitHub repository https://github.com/HammadShaikh-051/Jira-Experiment in the web browser.
2. Edit a repository file or push changes from a local terminal.
3. Construct a standard Git commit message incorporating the Jira issue key and the Smart Commit workflow transition tag:

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SCRUM-5 Add WhatsApp chat integration #done

4. Commit directly to the main branch.

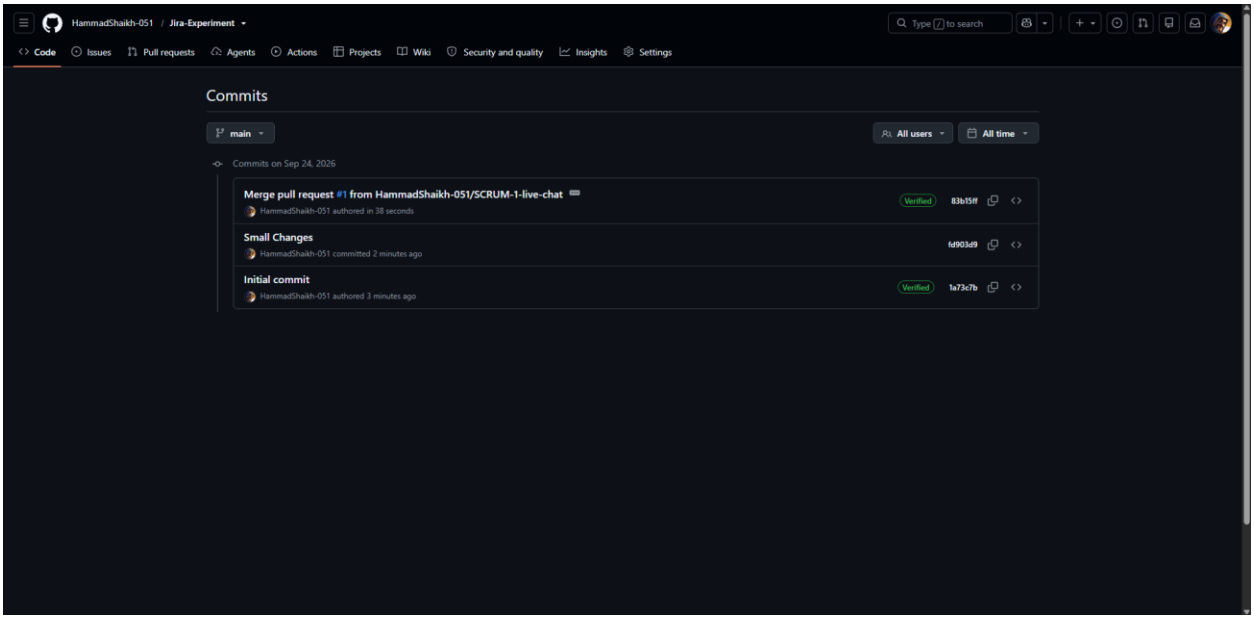


Figure 5A: GitHub repository commit history displaying the commit message referencing issue key SCRUM-5 *and the* #done *smart command.*

Step 6: Verification of Live Reflection in Jira

1. Switch back to the Jira project board and refresh the browser interface.
2. Open issue SCRUM-5 (Add WhatsApp chat to our website).
3. Verify the **Development** section in the right-hand details panel:
    - It displays **1 commit** linking directly to the Git commit hash in the https://github.com/HammadShaikh-051/Jira-Experiment
4. Verify the issue status:
    - The status has automatically transitioned from **In Progress** to **Done** via the #done Smart Commit directive without manual board updates.

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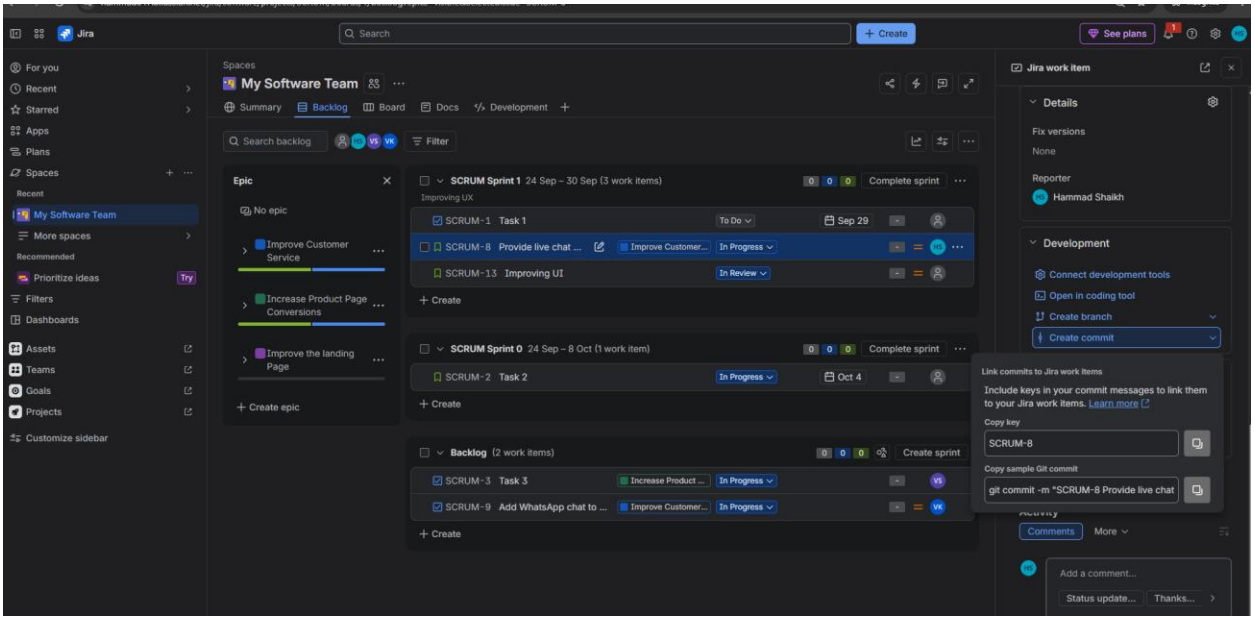
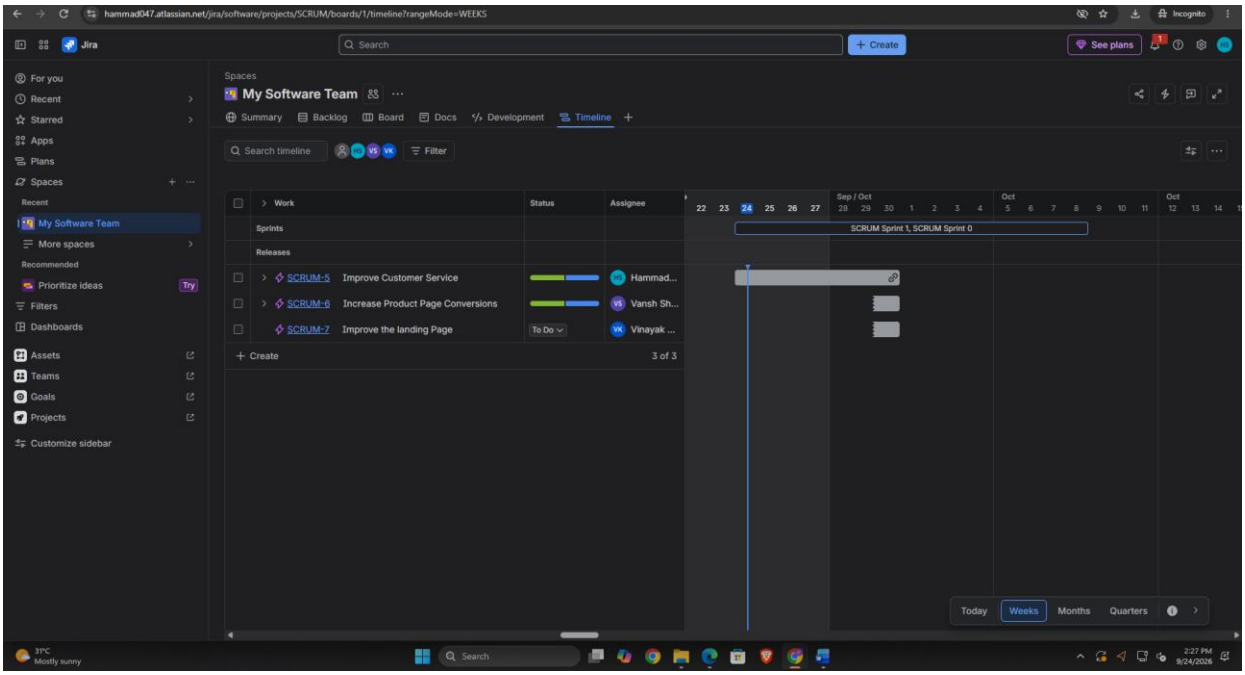


Figure 5B: Jira Work Item Details pane displaying automated status transition to Done and linked commit in the Development panel.



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